

Environmental Protection Acts (EPA)

1.Introduction

In this modern world a major part of innovation in scientific and technological development has been directed towards generations of elevation of human comforts, thereby increasing the standards of living in the society. As a result there is an enormous increase in industrialization. The improvement in the standards of human life is with respect to the applications of science and technology which includes;

- a) Production of more and better quality of food.
- b) Innovations in communication systems.
- c) Creation of reliable and faster transportation.
- d) Need of safe water.
- e) Providing healthy environment.
- f) Protecting the environment from natural disasters ex.droughts,floods,volcanic eruptions,etc.

As a result these major needs by the growing mankind led to the disturbing side effects such as environmental pollution, deforestation, urbanization, loss of arable land etc. Hence, protecting our earth and environment from all these effects are the responsibility of each and every individual. In addition to this it is also the obligation of the State and all the other State organs including courts.

In this regard, environmental protection acts has succeeded in unshackling man's right to life and personal liberty from the clutches of common law of individual ownership. The judiciary in our country has been exhibiting exemplary concern and appreciation towards environmental protection and ecological conservation.

1.1 Environmental legislation in India

Our country has demonstrated its concern for pollution control and environment protection by enacting several legislations and constituting Statutory Bodies dedicated for this cause. The important acts that had been put forwarded regarding the environmental protection are as follows;

- 1) Air act
- 2) Water act
- 3) Forest conservation act
- 4) Wildlife protection act

The environment where we live, there is an effective interaction of mankind with its components like water, air and land. The human activities with the ecosystem results in deteriorating phase A on account of the overexploiting the natural resources. Hence it is needed that every individual, Institutions, and Organisations along with the Government should come forward to protect the environment where we live.

This was first discussed in the “United Nations Conference on the Human Environment” , held in Stockholm on 5th June 1972, during which the various issues related to environmental pollution was discussed in detail by the participating countries. India was one among them and raised its concern regarding the environmental protection.



The outcome of the conference resulted in declaring various guidelines mentioning the role of individual and the Government in safeguarding the natural resources. To emphasis the decisions taken and also to signify the importance of protecting the environment 5th June is observed as “World Environment Day” every year.

2.The Environment (Protection) Act

One of the important outcomes of this event is, “**The Enviromment (Protection) Act**” put forward in the year 1986. The act was passed under Article 253 of the Constitution, and came into force 19 Nov 1986- birthday of Late Prime minister, Indira Gandhi who was one of the world's first leaders to embrace policies to protect the environment, long before it became a survival imperative. This was a welcoming venture in the field of environmental legislation which was a long –felt need. This act provides overall protection and preservation of the ecosystem. In our country this was enacted after the Bhopal gas tragedy. The Act consists of 26 sections and divided into four chapters. This relates to some of the preliminary aspects which have to be concentrated on. This speaks about the general powers of the central government.

This also includes the prevention control, and abatement of environmental pollution to prevent the contamination and also how to avoid the pollution due to air water as well as the other pollutants.

The four chapters formulated address the following;

Chapter-I: Related to preliminary information and definitions on Environment, Environmental Pollutant, Environmental Pollution and Hazardous Substance.

Chapter-II: Explains the powers of the Central government regarding the protection and improving the quality of the environment.

Chapter-III: Prevention, control and abatement of environmental pollution

Chapter-IV: Miscellaneous; Protection of action taken in good faith.

2.1 Objectives of EPA

-  To Provide protection and improvement
-  To Prevent environmental pollution
-  To tackle specific environmental problems in different parts of the country
-  To coordinate activities of various regulatory agencies already in existence
-  To appoint environmental officers to check for pollution
-  To improve the quality of life by protection of environment
-  To establish environmental laboratories and to monitor the pollution level at various parts of the country.
-  To protect forests and wildlife in the country

2.2 Environmental Protection Acts Need:

We have been witnessing a decline in the quality of the environment, and we are evidencing that this is due to the increase in the population, which is resulting in environmental pollution and because of which there is heavy loss of vegetation as well as the biodiversity. There is also excessive chemical concentration that is happening, which is also contributing to the pollution and growing risk of environmental accidents.

Example: Bhopal gas tragedy.

This was proposed in 1986 after witnessing the consequences of the Bhopal tragedy. This is supposed to be one of the worst industrial disasters in history, which took place in 1984 from December 3rd to 4th. This was due to the chemical compound abbreviated as MIC, which is methylisocyanate. It was released from the Union Carbide plant at Bhopal. This gas leakage led to various environmental issues. After witnessing the impacts of this particular accident, the Act was passed severely, and that came into the picture, which is called the Environmental Protection act. This Act was passed under article 253 of the Constitution and came into force on November 19th, 1986. To honor Mrs. Gandhi's birthday, November 19th, that particular Act was passed on November 19th, 1986. This Act acts as umbrella legislation, which is designed to provide a framework for central government and also to coordinate the various activities of various Central as well as state authorities.

2.3 Provisions of the Environmental Protection Act:

It lays the standards for the quality of air, water, and soil. This comes out with the maximal permissible limit of the pollutants. Procedures to handle and safeguard the hazardous substance were formulated. Initiation has taken place to identify some areas that are called restricting.

In the area where the hazardous materials are handled, the disposal and treatment of the hazardous material can be treated by throwing the waste into those restricting areas, and the required process can occur. There is the involvement of both pollution control boards, the State and the central boards to monitor the pollution level and also to prevent and avoid the accidents which might be causing the pollution:

2.4 Role of pollution control boards:

- Advise the industries about the treatment of effluents and using Advanced Technologies.
- Emphasize the importance recycle and reusing then reducing waste.
- To encourage the recovery of biogases, to make the people understand the importance of energy and how to convert the materials into reusable materials.
- Importance to implementation of cleaner Technologies in the industries.

Example: leather tanneries-the leather processing units are set up in the various parts of the State, especially in the Tamil Nadu

2.4 Critical definitions

Environment

It includes the major components like water, air, land, and interrelationships among them and between the living beings.

Environmental pollutant

This could be any substance like gas, liquid, or solid substances, which could also be a chemical compound. It is supposed to be present in the environment beyond its permissible limit, an environmental pollutant.

Environmental pollution

Environmental pollution is nothing but the process in which the pollution is happening.

Several sets of Rules relating to various aspects of management of hazardous chemicals, wastes, micro-organisms etc. have been notified under this Act.

2.5 Some important features of this Act:

- The Central Government may also put restrictions on an area in which any industry, operation or process or class of industries or operations shall not be carried out. If they are to be carried out, they may be permitted with certain safeguards.
- Emissions and effluent standards in respect of 61 categories of industries have been evolved notified so far.
- The standards in respect of pollutants are to be achieved within a period of one year from the date of their notification, especially by those industries identified as highly polluting.
- However, if a particular SPCB may so desire, it may reduce the time limit and also specify more stringent standards in respect of a specified category of industries within their jurisdiction. The SPCB however cannot relax either the time limit or the standards.
- Those industries that require consent under the Water Act, Air Act or both, or authorization under the Hazardous Waste (Management and Handling) Rules, 1989, are

required to submit an environmental audit report to the concerned SPCB/PCC on or before 30 September every year.

3. The Water (Prevention and Control of Pollution) Act, 1974

The pollution of water is an important fact which was felt by each citizen and this was brought as a major problem faced all over the world. To address the pollution of water due to various activities caused by the growing industries and the excess use and exploitation of the ground water conservation of water and its optimal use is being insisted. To address the pollution and prevention of water the Water Act was enacted by Parliament in 1974 with a focus on the prevention of control of water pollution and the maintaining or restoring of wholesomeness of water. The act is applicable in all the states of India. Some salient features of this Act are as follows:

- No person shall knowingly cause or permit any poisonous, noxious or polluting matter determined in accordance with such standards as may be laid down by the State Board to enter (whether directly or indirectly) into any stream or well or sewer or on land;
- No person shall knowingly cause or permit to enter into any stream any other matter which may tend, either directly or in combination with similar matters, to impede the proper flow of the water of the stream in a manner leading or likely to lead to a substantial aggravation of pollution due to other causes or of its consequences.
- No person shall, without the previous consent of the State Pollution Control Board (SPCB):
 - a) establish or take any steps to establish any industry, operation or process, or any treatment and disposal system or an extension or addition thereto which is likely to discharge sewage or trade effluent into a stream or well or sewer or on land, or
 - b) Bring into use any new or altered outlets for the discharge of sewage, or
 - c) Begin to make any new discharge of sewage.

The act also advises the investor intending to set up an industry is required to apply in the prescribed form to the SPCB concerned to obtain the consent to establish as well as the consent

to operate the industry after establishment. While granting the consent, the SPCB also stipulates specific conditions relating to the temperature, volume, composition, rate and point of discharge emissions, effluents, etc. The consent to operate an industry is granted for a specific period after which the conditions stipulated at the time of granting consent are reviewed by the SPCB. Even before the expiry of the consent period, the SPCB is authorized to carry out random checks on any industry to check if the standards prescribed are being complied with by the industry; in case the standards are not being met, the SPCB is authorized to serve a notice to the concerned person. The owner of a defaulting industry may be required to construct a sewage/effluent treatment system. In the event of non-compliance of the standards, the SPCB may issue directions for disconnecting electricity and water supply or any other services to the industry, in extreme cases even to close down the unit. Stringent penalties are prescribed in this Act for those who operate their industry without the valid consent or in violation of consent conditions. Any person aggrieved by an order of the SPCB in the above context may appeal to the Appellate Authority constituted by the concerned State Government. As per the provisions under the Water Act and the Air Act, all state governments, are required to constitute Appellate Authorities for addressing the appeals received against SPCB.

3.1 Salient features of the water Act

- Provide the prevention control and abandonment of water pollution.
- Maintenance or restoration of the wholesomeness of the water.
- It is designed to assess the pollution level and punish the polluters
- Central as well as the State Board has been set up to monitor the water pollution and is expected to provide the guidelines.

3.2 Roles of central Pollution Control boards

- advise the central government regarding the corrective measures that have to be taken
- coordinate with the state government to provide technical guidance and both the pollution control boards
- organizing some training programs where in which they can Define the functions of various authorities

- create awareness through the different Communications media and Technical and statistical data collection
- come up with some manuals which could address the treatment of sewage as well as the industrial effluents
- establish the standards for the water quality parameters
- develop and recognize labs for the analysis

3.4 Roles of State pollution-controlled boards

- Advise the state government regarding the corrective measures to be taken if the water bodies are severely polluted,
- Empowered to take samples from various water bodies and give the guidelines that they can monitor the water level to what extent it is contaminated.
- Powers to close down any Industries if they are crossing the permissible limit
- and any new Industries if the people want to set up, then they are supposed to get the concern from the state pollution control board

example: standard effluent treatment plant (ETP). It has been installed near ranipet in Vellore to address the effluent problem that is cast from the various tanneries Industries.

4. The Air (Prevention and Control of Pollution) Act, 1981

The act is one of the outcomes of the International Conference where the environmental protection was discussed. The air act speaks about the prevention and control of the air. This addresses the pollution that is due to the solid-liquid and gaseous substances.

This consists of 50 sections, and it was amended in the year 1987, where noise pollution was also inserted.

The objective of the Air Act is to prevent, control and reduce air pollution including noise pollution and to establish Boards at the State/UTs for this. Under the provisions of this Act, no person shall establish or operate any industrial plant without the consent of the SPCB/ PCC Industry, Vehicles, power plants, Etc are not permitted to release particulate matters which might be including lead, carbon, carbon monoxide, sulfur dioxide, and nitrogen oxides, as well as volatile organic compounds or other toxic substances beyond a prescribed level.

This is an Act to provide for the prevention, control and abatement of air pollution, for the establishment of boards with a view to carrying out the air pollution control and for conferring on and assigning to such Boards powers and functions relating thereto and for matters connected therewith. It is considered necessary to implement the decisions aforesaid in so far as they relate to the preservation of the quality of air and control of air pollution..

For obtaining consent to establish an industry, the investor has to apply to the SPCB/PCC in the prescribed form accompanied by the prescribed fees. The Board is required to grant consent within 4 months of receipt of the application. The consent would contain conditions relating to specifications of pollution control equipment to be installed. Other Provisions of the Act are similar to those of the Water Act.

5. The Forest (Conservation) Act 1980

The forest is supposed to be a natural resource that is available abundantly. The earth is covered these forest resources, which will be seen as green in color, called the green blanket of the world. In addition to this, it also has various value-added products. They are classified into different categories like ecological significance products, commercial products, environmental applications, and societal applications. It is also a natural source of medicinally important compounds.

It is also identified as a home for wildlife. It provides wildlife habitat. There are about seven million species that depend on this forest. Forest also contributes to the regulation of the hydrological cycle in which 50 to 80 of the moisture in the air is transformed into various transpiration processes.

It reduces global warming by absorbing the carbon dioxide gas from the atmosphere, contributes to soil conservation where the soil is tightly packed. Protection and conservation of forest and its resources are the important requirement to protect the mother earth.

Objectives

1. To protect and conserve the forest.
2. To ensure judicious use of forest products

5.1 Salient features

- To protect and the forest from illegal cutting, encroachments, fire, grazing and shift agriculture
- To prevent deforestation activities
- The reserve forest should not be de-reserved without the prior permission of the Central Government.

6.The Wildlife Protection Act 1971(Amended in 1982, 1986, 1991, 1993, 2002 and 2006)

This act was enacted for providing protection to wild animals and birds. It provides for establishment of sanctuaries and national parks. It has provisions for dealing with zoos, trade in wild animals and for taking action for the specific protection of certain species. The act also provides for the constitution of a Wild Life Advisory Board, appointment of Chief Wild Life Warden and other employees by the State Governments for the protection of wildlife. It has been observed that “The Act failed to frighten poachers and miners, but was used against animal and environment-friendly tribals”.

The Act resulted in establishing the following national parks, Sanctuaries,conservation reserves and biosphere reserves in our country.

- 104 National Parks,
- 551 Sanctuaries,
- 86 conservation reserves and
- 11 biosphere reserves in India.
- Endangered Wildlife list is created and updated every year.

7. The Environmental Impact Assessment

The following are the two distinct stages in EIA.

1. Preliminary Assessment: Carried out in the early stage of planning.
2. Detailed Assessment: Carried out during project planning until the project plan is completed and is reported formally as Environmental Impact Statement.

7.1 The key elements of an EIA are the following:

- 1) Screening
- 2) Scoping
- 3) Impact ID and prediction
- 4) Impact assessment and mitigation
- 5) EIA report
- 6) Decision making
- 7) Monitoring and evaluation

It has to be followed when a project or a policy has been proposed

Scoping is used to identify the key issues of concerns at an early stage in the planning process. . The scoping process should involve all interested parties such as the proponent and planning or environmental agencies and members of the public. The results of scoping will determine the scope, depth and terms of reference to addressed within the Environmental Statement. The following are the major objectives of the scoping.

- To identify the key issues and concerns of the interested parties
- To identify who is concerned
- To identify what their concerns are
- To identify why they concerns are
- To identify what is the threshold of concern where change becomes unacceptable.

Screening

This is used to decide whether an Environmental Assessment is required. There are two lists, referred to as schedules.

Schedule 1 – Environmental Assessment are required in every case. Schedule 1 projects range from “an integrated works for the initial melting of castiron and steel” to “a thermal power station or other combustion installation with a heat output of 300 MW or more”.

Schedule 2 – Environmental Assessment are required if the project is likely to give rise to significant environmental effects by virtue of factors of their nature, size or location. The list of Schedule 2 project is greater than that of Schedule 1.

Screening is comprehensive and clear method of decision making. It is practical, quick and easy to use.

The other process like Impact ID and prediction, Impact assessment and mitigation insists about the steps involved in the implementation of the project or policy.

This would include the various pollution abatement techniques that would be required to reduce emissions to the permissible limits. If the uncertainties are great, with the possibility of grave consequences and no mitigating measures, then the development plan is rejected. If there are uncertainties that might be reduced by further studies then the applications deferred until further studies are carried out.

7.2 Environmental Statements/Reports

The EIA is the process required to produce the Environmental Statement. The environmental impact statement is a comprehensive document that reports the findings of the EIA. This is the final stage of the EIA process and is now often required by law before any new project can proceed.

Once if all the requirements are met with then the decision making will be made by the officials involving State and Central Board. However, once the project is approved continuous monitoring is emphasized to address the requirements of the environmental protection act.

7.3 Methodologies for EIA

There are various methodologies or tools available to assess the impact of any activity that can be approved. It includes;

1. Ad hoc method- Simple
2. checklist methods
3. matrix method
4. network methods
5. overlay methods

These are the significant methodologies that are proposed to implement the environmental impact assessment. The best method should be able to

- Organize a large mass of heterogeneous data
- Allow summarization of data
- Aggregate the data into smaller sets with least loss of information
- Consider the target audience

7.3.1 The first method is known as Ad-hoc approach, this is straightforward method all people will be able to understand. There are various characteristics that have been included like wildlife, the species, the natural vegetation and the groundwater quality, noise pollution, air quality grazing, and also the other component that are needed to be considered before that particular project could be implemented. It gives a clear idea regarding the stages as well as the parameters that has to be considered before bringing in that particular method.

It could be done by the opinion poll method, expert opinion method, and Delphi method.

The opinion poll method is based on the opinions that the various stakeholders. We can get the information through taking a survey to get the details on particular project will be effectively implemented in that particular area and the second one is the expert opinion. When the proposal and policy have been drafted it will be given to an expert. The particular policies or the

proposals that has been submitted will be given to the experts to evaluate the project. There will be an opinion that is obtained by an expert and based on which the method can be approved.

The third one is the Delphi method, which will be a questionnaire which consists of questions that will be distributed among the various stakeholders to gather the information pertaining to the project proposed. The responses will be analyzed seriously, and then that particular project can be implemented. The advantage is that it is very simple and easily understandable and it is a draft method that will give us information about the impact on the environment.

The disadvantages of Ad hoc method:

- It requires expertise.
- It is not applicable to large data.
- It is on a guest basis.
- It gives only minimal guidance
- It does not allow us to address the actual impacts on the specific parameters that will be affected.

7.3.2. The second method is known as the checklist method.

This checklist method lists the various environmental factors in a structured format. The checklist method can be classified into four different types

- Simple – list of environmental factors with no guidelines
- Descriptive- includes the guidelines and identification of parameters.
- Scaling- similar to checklist method but with the additional information on parameters.
- Weighing – similar to scaling but with additional information for the evaluation of each parameters.

Example:

Resources	Potential Impact
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	Site clearance	Laying pipelines	Treatment process	Import materials
Air Quality	✓			
Water quality			✓	
Floara		✓	✓	
Founa			✓	
Noise				✓

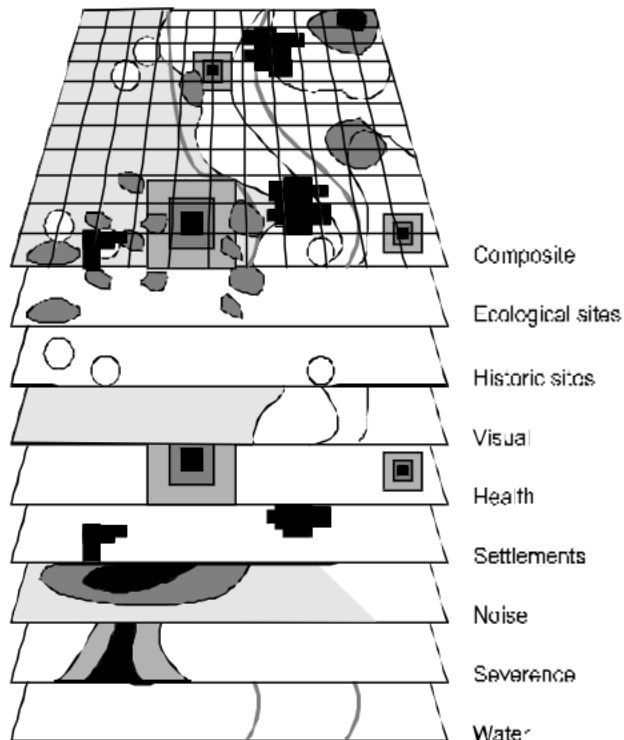
7.3.3 The third method is Matrix method. This matrix method involves a two-dimensional checklist in which one axis represents the project activities, and the other can represent the environmental characteristics. It is a 2-Dimensional check list where one axis represents project activities and the other represents environmental characteristics. It can evaluate the degree of impact on the environment. It can evaluate the cumulative and indirect impacts and interaction with natural resources.

Example: Leopold Matrix Method

Fourth method is known as Network method. This method

- Identifies the pathway using series of network
- Integrated assessment –integrates all aspects
- It extends to include both the primary as well as the secondary impacts.
- It is shown in reference or sequence diagram.
- Identification of direct, indirect along with short, long term impact is a crucial and basic step of making an impact tree

7.3.5. Finally the overlay method involves set of transparent map.



It displays the spatial distribution of environmental characteristics. It can display both physical and social aspects of area under study. The impacts of the project are well predicted before and after implementation of the project. This involves advanced techniques like GIS technique which is used to predict and study the project area and assessments accurately.

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